F.COM GROUP PROJECT

Presented by TEAM 5

TOPIC: STRING UTILITY BOX

TEAM 5 INCLUDES:

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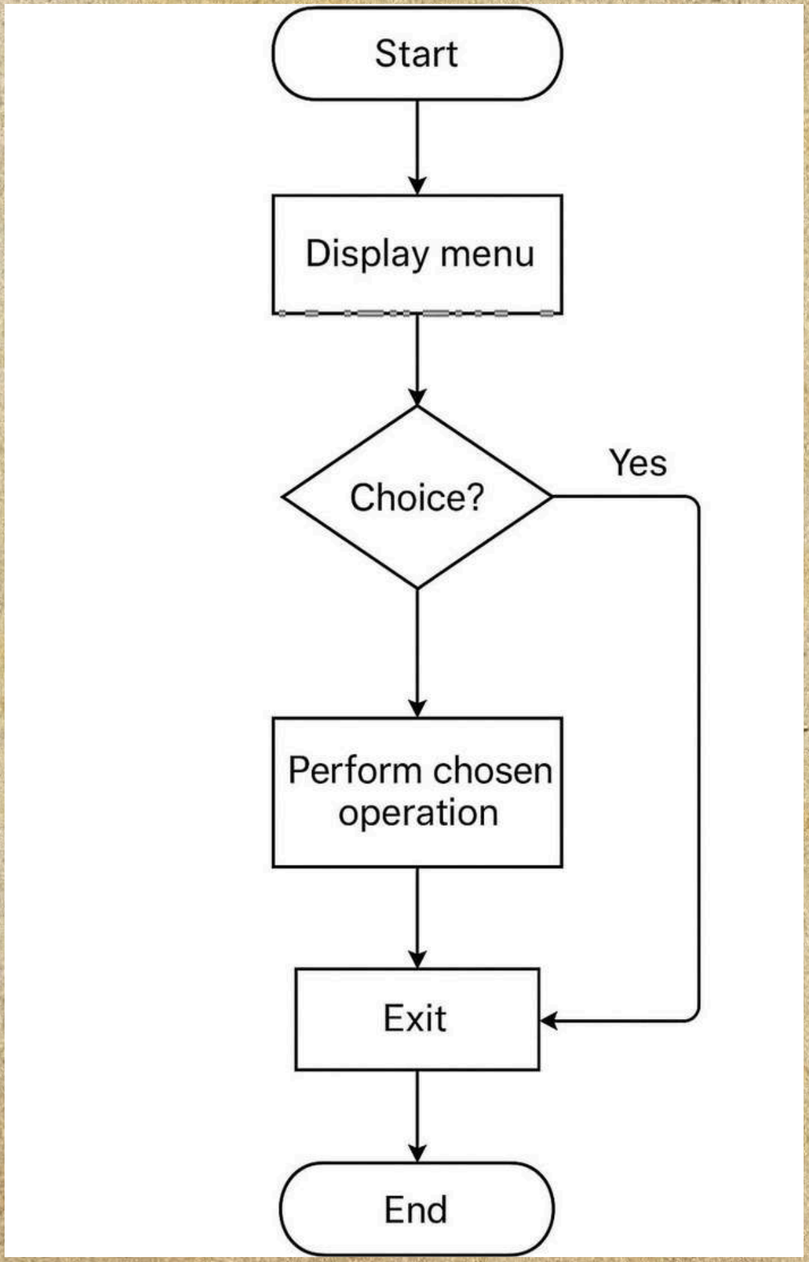
FLOW CHART


Start
Display menu
Choice?
Yes
Perform chosen operation
Exit
End

OBJECTIVE AND PROBLEM STATEMENT

*OBJECTIVEOFTHESTRING UTILITY BOX PROGRAM:
TO CREATE A MENU-DRIVEN C PROGRAM THAT PERFORMS VARIOUS STRING OPERATIONS SUCH A
FINDING LENGTH, REVERSING, CONCATENATING, COMPARING, CHANGING CASE, AND COUNTING
WORDS — ALL WITHIN A SINGLE INTERACTIVE TOOL FOR EASY STRING MANIPULATION AND
UNDERSTANDING OF STRING FUNCTIONS IN C.

THE PROGRAM SHOULD LET THE USER CHOOSE FROM OPTIONS SUCH AS FINDING THE LENGTH OF A STRING, REVERSING IT, CONCATENATING TWO STRINGS, COMPARING TWO STRINGS, CHANGING THE CASE (UPPERCASE/LOWERCASE), AND COUNTING THE NUMBER OF WORDS

MODULES

1.MAIN MODULE – DISPLAYS THEMENU, GETSUSER CHOICE,AND CALLS THE REQUIRED FUNCTION. 2.LENGTH MODULE – FINDS AND DISPLAYS THE LENGTH OF THE STRING. 3.REVERSE MODULE – REVERSES THE GIVEN STRING. 4.CONCAT MODULE – CONCATENATES TWO STRINGS. 5.COMPARE MODULE – COMPARES TWO STRINGS AND SHOWS THE RESULT. 6.UPPERCASE MODULE – CONVERTS ALL CHARACTERS TO UPPERCASE. 7.LOWERCASE MODULE – CONVERTS ALL CHARACTERS TO LOWERCASE. 8.WORD COUNT MODULE – COUNTS THE TOTAL NUMBER OF WORDS IN A STRING. 9.EXIT MODULE – ENDS THE PROGRAM.

CODING

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main(){
    char s[200], t[200]; int c;
    printf("Enter string:\n"); fgets(s,200,stdin); s[strcspn(s,"\n")] = 0;
    do{
        printf("\n1.Len 2.Words 3.Rev 4.UP 5.low 6.Exit\nChoice: ");
        if(scanf("%d",&c)!=1) break; getchar();
        switch(c){
            case 1: printf("Len=%d\n",(int)strlen(s)); break;
            case 2: { int w=0,i; for(i=0;s[i];i++) if(!isspace((unsigned char)s[i]) && (i==0||isspace((unsigned char)s[i-1])))
printf("Words=%d\n",w); } break;
            case 3: { int i; for(i=(int)strlen(s)-1;i>=0;i--) putchar(s[i]); putchar('\n'); } break;
            case 4: strcpy(t,s); for(int i=0;t[i];i++) t[i]=toupper((unsigned char)t[i]); printf("%s\n",t); break;
            case 5: strcpy(t,s); for(int i=0;t[i];i++) t[i]=tolower((unsigned char)t[i]); printf("%s\n",t); break;
        }
    } while(c!=6);
    return 0;
}
```

RESULT

```
Enter string:
jeyasingh

1.Len 2.Words 3.Rev 4.UP 5.low 6.Exit
Choice: 1
Len=9

1.Len 2.Words 3.Rev 4.UP 5.low 6.Exit
Choice: 2
Words=1

1.Len 2.Words 3.Rev 4.UP 5.low 6.Exit
Choice: 3
hgnisayej

1.Len 2.Words 3.Rev 4.UP 5.low 6.Exit
Choice: 4
JEYASINGH

1.Len 2.Words 3.Rev 4.UP 5.low 6.Exit
Choice: 5
jeyasingh

1.Len 2.Words 3.Rev 4.UP 5.low 6.Exit
Choice: 6

--------------------------------
Process exited after 20.5 seconds with return value 0
Press any key to continue . . .
```

OUTPUT EXPLANATION

THE PROGRAM ASKS THE USER TO ENTER A STRING.
IT THEN SHOWS A MENU (1–6) FOR DIFFERENT STRING OPERATIONS.
BASED ON THE CHOICE, IT DISPLAYS:

1 FIND THE LENGTH OF THE STRING

2 READING AND CONCATENATING THE STRINGS

3 COMPARING TWO STRINGS

4 CHANGE THE CASE (LOWER/ UPPER)

5 EXIT THE PROGRAM

CONCLUSION

THE PROGRAM EFFICIENTLY PERFORMS MULTIPLE STRING OPERATIONS USING A SIMPLE AND USER-FRIENDLY MENU SYSTEM.

THANK YOU!!!

PresentedbyTeam 5